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Innovation and internationalization through exports

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Abstract
Successful product innovation leads to the decision by small and medium enterprises (SMEs) to enter the export market. We argue that in addition to a direct effect of innovation on exports, product innovation, through its effect on firm productivity, increases the likelihood of the firm entering the export market. Using a panel of Spanish manufacturing firms, we show that the strong positive association found between firm productivity and exports in the literature relates to the firm's earlier innovation decisions, and that, when controlling for product innovation, the relationship between productivity and exports vanishes for these innovating firms.
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Keywords: exports; innovation; product innovation; process innovation; productivity

INTRODUCTION

In the process of internationalizing, firms typically move through different stages over their lifetime, starting from exporting operations and advancing to other forms of international involvement, such as foreign direct investment by acquisition, greenfield investment or joint venturing (e.g., Johanson & Vahlne, 1977, 1990; Vernon, 1966). Exporting is a particularly important internationalization strategy for small and medium-sized firms (Leonidou & Katsikeas, 1996). Compared with foreign direct investment (FDI), exporting is a relatively straightforward way of entering foreign markets, because it involves comparatively low levels of commitment and risk, and the firm does not have to deal with the complexities of establishing a foreign subsidiary (e.g., Lu & Beamish, 2006). Therefore, exports constitute the initial preferred way of internationalizing for small and medium enterprises (SMEs) (Young, Wheeler, & Davies, 1989).

The decision of a firm to enter international markets is in itself a sequential process. Vernon (1966) hypothesized about how the product life cycle and the firm's initial innovation outcomes drive the internationalization process of products and firms. In a first step the firm innovates through the generation of a product based on some opportunities that are detected in the home market of the firm. As demand for the product starts to appear elsewhere, the firm decides to internationalize by exporting this product to markets with demand for products with similar characteristics. Vernon (1979) argues that such a sequential internationalization process based on product innovation characterizes the

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internationalization of innovative SMEs particularly well. These SMEs may not have sufficient resources and capabilities to innovate for global markets, and so start from the home-based innovation, moving gradually to the possibility of exporting and eventually to FDI.

In this paper, we investigate the drivers of the exporting phase in the internationalization process of firms. Prior research found innovation, and more specifically product innovation, to be an important factor in explaining the entry into the export market (e.g., Basile, 2001; Becker & Egger, 2009; Cassiman & Martinez-Ros, 2007). We distinguish between two different channels through which (product) innovation can affect the decision of a firm to start export operations. On the one hand, successful innovation may serve as a productivity-enhancing investment, hence leading indirectly to the self-selection of firms into export markets. On the other hand, innovation activity may also have a direct effect on the export decision. Firms may start operations abroad in search for greater demand for their new products, or in order to spread out the research and development (R&D) costs over the larger sales volumes. In this study we are particularly interested in the indirect productivity channel effect of (product) innovation on the decision to start exporting, and in determining the relative quantitative contribution of this effect and, as a result, take a first step towards explaining the well-documented productivity – exports association. We test the proposed relationship between innovation and the productivity – exports association using a long panel of Spanish manufacturing firms. Our findings suggest that product innovation is an important factor in explaining the export – productivity link. Once controlling for product innovation, the relationship between productivity and exports vanishes for innovators.

Our paper contributes to the literature in several ways. First, in developing our argument on the internationalization process of SMEs, it provides a logical connection between three distinct literatures without much overlap, but dealing with very related questions. From the international business literature we draw on explanations of the patterns of internationalization for newly developed products for smaller firms (Chang, 1995; Vernon, 1966, 1979). From the studies on productivity and exports and learning-by-exporting we build on the selection mechanism of relatively more productive firms into the export market as a dominant explanation for the export – productivity correlation (e.g., Clerides, Lach, & Tybout, 1998; Delgado, Farinas, & Ruano, 2002). Finally, we draw on the studies of productivity determinants that find that R&D and innovation activities are key drivers of productivity heterogeneity across organizations (e.g., Bartelsman & Doms, 2000; Griliches, 1998).

Second, we contribute to the literature on the productivity – exports link by empirically showing the importance of prior firm investments such as innovation for explaining the self-selection mechanism of more productive firms into exports. Our results highlight the importance of firm decisions for their future growth paths, and for explaining performance heterogeneity among firms. Third, this study adds to our understanding of the export strategies of SMEs. In particular, prior empirical research on the antecedents of the export decision highlighted the importance of product innovation for the decision to export (Cassiman & Martinez-Ros, 2007). We provide evidence on the existence of two distinct channels for the effect of product innovation on exports.

The paper proceeds as follows. In the next section we discuss the background literature and theory leading up to the empirical prediction we propose to test. The third section describes our sample, and our statistical approach. The fourth section presents the results of the empirical analysis. The final section provides a discussion of the results, and offers some conclusions and directions for future research.

INNOVATION, PRODUCTIVITY AND EXPORTS

In the empirical international trade literature, the positive association between exports and firm productivity has been well documented. At least two explanations for the observed export – productivity link have been suggested: see, for example, Wagner (2007) and Greenaway and Kneller (2007) for a survey of the literature. On the one hand, the positive association between exporting and productivity is explained through a selection mechanism, whereby the more productive firms enter into the export market. On the other hand, there is the possibility of learning-by-exporting – exporters may learn from their foreign contacts, adopting new production technologies and enhancing productivity and performance (Salomon & Shaver, 2005).

With both mechanisms being plausible, empirical evidence for the developed countries is rather unanimous in supporting the selection hypothesis

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behind the export – productivity association. Sunk start-up costs associated with becoming an exporter lead to the self-selection of more productive firms into exporting. The hysteresis in exporting serves as evidence for the sunk entry costs in the export market (Campa, 2004). The general finding in developed countries is that exporting firms have higher productivity than non-exporters before taking up exports, and no significant productivity advantages are observed among continuous exporters or non-exporting firms respectively over time (Aw, Chen, & Roberts, 2001; Bernard & Jensen, 1999; Clerides et al., 1998; Damijan & Kostevc, 2006; Delgado et al., 2002; Fafchamps, El Hamine, & Zeufack, 2007; Greenaway & Kneller, 2007; Roberts & Tybout, 1997).[1]

Such heterogeneity in productivity before entering into the export market raises an important question about the sources of high productivity of these exporting firms. How do firms obtain higher productivity levels that allow them to enter the export market, effectively setting off the internationalization process of the firm? International trade literature, following the work on industry dynamics (Hopenhayn, 1992; Jovanovic, 1982), has attempted to incorporate firm heterogeneity in the theoretical models. These theoretical models on industry dynamics have assumed that firms are born with an inherent ability – their productivity. Efficient firms survive and grow in the market, while inefficient ones, with productivity below a certain threshold, decline and fail (Hopenhayn, 1992; Jovanovic, 1982). Most of these models, however, assume that the productivity distribution across firms is exogenous to firms, thus relating firm survival to the luck of the draw. Firms with low productivity exit, while "lucky" firms with high productivity survive and continue growing. Little room is left for firm decisions, except for the decision on exiting. Theoretical work by Melitz (2003) and Bernard, Eaton, Jensen, and Kortum (2003) formulates theories that reflect the empirical regularities observed in exporting behavior and productivity. These models, however, do not explain why these firms are more productive and self-select into exporting: that is, the theories are not causal theories between firm decisions and their subsequent decision to export – something that would be important to explain to interest international business scholars. An exception is recent work by Costantini and Melitz (2008). They explore the link between investments in innovation and the decision to export in the context of the liberalization of trade regimes, and relate innovation to a shock in future productivity. Similarly, Lileeva and Trefler (2010) show for a sample of Canadian firms that the decisions to export and to invest in productivity (e.g., by investing in product innovation) are positively related, and may be complementary for productivity growth.

One important source of productivity differences in general is argued to be related to R&D and innovation investments made by the firm (see Griliches, 1998). These studies on the effect of R&D and innovation on productivity at the firm level find that R&D investment and innovation activity indeed constitute an important source of the observed productivity differences between firms (e.g., Crepon, Duguet, & Mairesse, 1998; Doraszelski & Jaumandreu, 2007; Griffith, Huergo, Mairesse, & Peeters, 2006; Huergo & Jaumandreu, 2004). Crepon et al. (1998), estimating a structural model that links productivity, innovation output and innovation inputs, find that firm productivity correlates positively with higher innovation output. Huergo and Jaumandreu (2004), using a panel of Spanish firms, find that process innovation is an important determinant of productivity growth at the firm level. Investigating the relationship between innovation and productivity in four European countries, Griffith et al. (2006) find – consistent with the previous studies – that both product and process innovations have a positive, significant effect on firm-level productivity in three out of the four countries. While some of the earlier studies did distinguish between product and process innovation without any clear different effect, Foster, Haltiwanger, and Syverson (2008) find that firm-specific demand shocks rather than production efficiency shocks seem to account for more of the difference in measured firm productivity, consistent with a larger effect of product rather than process innovation on productivity.

At the same time, R&D and innovation activities play an important role in explaining a firm's decision to export and export volumes. Vernon (1966, 1979) hypothesized about the firm's internationalization process following the product life cycle. Young firms that possess a newly developed product in the early phase of the product life cycle based on their proprietary knowledge will start moving into exports to exploit their market power in foreign markets because of the limits the domestic market poses in the early growth stage (Hirsch & Bijaoui, 1985). Product innovation actually favors exports in the

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search for greater demand for these products of potentially superior quality (Hitt, Hoskisson, & Kim, 1997). In addition, exports contribute to increases in firms' sales, thus allowing the spreading of innovation costs, which to a large extent are fixed, over a greater demand volume (Alvarez & Robertson, 2004), while generating monopolistic advantages based on specific technological knowledge (Chang, 1995). Moreover, investments in innovation enable the firm to achieve greater ability to meet the demands of its changing domestic and international markets, thus making exporting more profitable for a firm (Zahra & Covin, 1994).

Recent empirical studies relate product innovation and exports directly. Basile (2001) shows, for a sample of Italian manufacturing firms, that firms introducing product and/or process innovations either through R&D or through investments in new capital are more likely to export. Bernard and Jensen (2004) find that firms switching primary SIC code, which could indicate new product introductions, significantly increase the probability of their entering the export markets. Furthermore, in a related paper, Cassiman and Martinez-Ros (2007) find a strong positive effect of product innovation, but not process innovation, on the decision of a firm to export. Becker and Egger (2009) find, consistent with our argument, that product innovation is of dominant importance relative to process innovation in explaining the decision of a firm to start export operations. Their results also show that process innovation matters for the decision to export only if accompanied by product innovation.

As a result, and based on the findings in these three streams of the literature on innovation, productivity and internationalization, we argue that (product) innovation increases the likelihood of internationalization of SMEs. First, R&D and (product) innovation are positively related with the firm's export decision through a direct demand expansion effect. However, there exists a second, more subtle channel through which innovation affects internationalization. R&D and (product) innovation positively affect firm productivity, and firms with higher productivity select into the export market. This second, indirect, channel suggests that product innovation may be responsible for both the productivity enhancement and the export orientation of a firm. Connecting product innovation, productivity and exporting, we argue that accounting for the previous innovation status of a firm might take us some way in explaining the observed positive association between exports and productivity. Hence, a potential underlying mechanism explaining the productivity improvements and consequently the decision of the firms to enter the export market is related to firms' innovation activity. Successful product innovation enhances the firm's productivity, leading to the selection of the more productive firms into the export markets.[2] In the following section we describe our data and methodology to test this expected relationship.

DATA AND METHODS

The Data

The data that are used in this study come from a survey (ESEE: Encuesta Sobre Estrategias Empresariales) of Spanish manufacturing firms started in 1990, with data collected annually up to 1998. The project was conducted by the Fundación Empresa Pública, with the financial support of the Spanish Ministry of Science and Technology. The information collected each year is consistent with the information in the previous years. The data include the population of Spanish manufacturing firms with 200 or more employees. They also contain a stratified sample of small firms constituting 4% of the population of small firms with more than 10 and fewer than 200 employees. Small firms that exited the original sample during the sampling period were replaced by firms with similar characteristics drawn from the population.[3] Previous research has used the same dataset, as it is representative for the Spanish manufacturing industry over this period (e.g., Campa, 2004; Cassiman & Martinez-Ros, 2007; Delgado et al., 2002; Salomon & Jin, 2008; Salomon & Shaver, 2005).

The ESEE dataset provides an appropriate setting to test the relationship between exports, innovation and productivity that we aim to examine. First, the data allow the firms and their export and innovation decisions to be traced over a time period of several years. Second, exporting firms constitute a large proportion in the sample and show considerable variation in their exporting behavior over time. Third, our data cover the period in the Spanish economy (the end of 1980 – 1999) characterized by opening towards the international markets. This makes our sample particularly well suited for examining the decision to internationalize. Fourth, there are very few firms (less than 0.3% of the sample) with FDI. In this way, we are able to focus on exports, without the

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confounding effects of other internationalization strategies. Although the percentage of firms engaged in FDI is less than 0.3%, about 10% of SMEs in the sample are foreign owned, that is, have more than 50% of foreign capital.[4]

Further, the sample contains detailed information on the firms' innovation behavior, and displays substantial variation across firms and over time. Finally, during 1990–1998 Spain had gone through an entire business cycle. In 1990 there was growth in the economy, followed by a sharp recession in 1993, and a recovery during the last years of the sample period (Shaver, 2007). Such variance can be usefully exploited to examine the productivity dynamics during 1990–1998.

The Sample of SMEs

In this paper we choose to focus on small and medium firms. We do so for several reasons. In the revised version of the product life cycle, Vernon (1979) suggested that, despite the changes in the global economy the hypothesized product life cycle behavior would still be relevant for smaller firms. These firms, unlike large companies, may not have enough resources or capabilities to innovate for global markets, and go global from the start, thus starting from their home-based innovation and moving gradually to the possibility of exporting and further to FDI.

We believe this to be particularly relevant for Spanish firms during the period of our sample. Spain may be described as a late-investor country (Pla-Barber & Escriba-Esteve, 2006). Before Spain joined the European Economic Community in 1986 it had little FDI abroad, being traditionally a net recipient of FDI. The situation has slowly changed with the opening of foreign markets through the entry into the European Community (1986), the establishment of a single European market (1993) and the adoption of the euro (1999). The favorable economic situation, coupled with public policies aimed at export promotions, encouraged the entry of firms into exports. Few firms do actually have any FDI in our sample. Only about 0.3% of the firms in the sample report FDI activities. Therefore, we believe that the sample of Spanish SMEs provides a good setting for testing the relation between innovation, productivity and exports. Moreover, product innovation may not be particularly important as a driver of exports for large firms, as these firms for the most part already export (and grew). About 81% of large firms in our data were exporters during the period 1990–1998, so we do not really observe the entry decision on the part of large firms. Nevertheless, the inclusion of large firms in the sample does not affect our results.

Recent globalization trends have influenced the patterns of internationalization of both large multinational enterprises and SMEs, leading to the emergence of so-called "born globals", that is, firms that from inception are active exporters, with a high percentage of sales from abroad (Oviatt & McDougall, 1994; Pla-Barber & Escriba-Esteve, 2006).[5] In our sample of SMEs the average percentage of sales coming from exports is 19% for the exporters, with 75% of exporters selling abroad less than 30% of their total output, which is not a characteristic of "born-global" companies.

The original sample of SMEs includes 1478 firms in 1990 and 1256 firms in 1998 from 20 distinct industries. Because of missing values the resulting sample is an unbalanced panel, with 8400 firm-year observations. Table 1 summarizes the export and innovation status of the firms in the final sample. Approximately 43% of the SMEs are exporters, with the proportion of exporting firms steadily increasing from 33% in 1990 to 52% in 1998. By contrast,

Table 1 Distribution of firms by year, 1991–1998

	1991	*1992*	*1993*	*1994*	*1995*	*1996*	*1997*	*1998*	*Total*
Percentage of exporting firms	0.33	0.37	0.38	0.42	0.47	0.47	0.51	0.52	0.43
	(0.46)	(0.48)	(0.48)	(0.49)	(0.49)	(0.49)	(0.50)	(0.50)	(0.49)
Percentage of firms with product innovation	0.20	0.22	0.21	0.21	0.22	0.20	0.22	0.21	0.21
	(0.40)	(0.41)	(0.41)	(0.41)	(0.41)	(0.40)	(0.41)	(0.40)	(0.40)
Percentage of firms with process innovation	0.27	0.25	0.27	0.28	0.25	0.24	0.28	0.30	0.27
	(0.44)	(0.43)	(0.44)	(0.45)	(0.43)	(0.42)	(0.45)	(0.46)	(0.44)
Percentage of firms with no innovation	0.63	0.64	0.62	0.62	0.63	0.64	0.60	0.58	0.62
	(0.48)	(0.48)	(0.48)	(0.48)	(0.48)	(0.48)	(0.48)	(0.49)	(0.48)
Number of observations	1011	1036	1055	1039	970	1029	1183	1079	8402

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Table 2 Transition probabilities (in %)

	No export (t)	*Export (t)*
No export ($t-1$)	91.7	8.3
Export ($t-1$)	8.4	91.6
	No product innovation (t)	*Product innovation (t)*
No product innovation ($t-1$)	89.3	10.7
Product innovation ($t-1$)	38.9	61.1
	No process innovation (t)	*Process innovation (t)*
No process innovation ($t-1$)	84.0	16.0
Process innovation ($t-1$)	42.9	57.1

the percentage of firms with innovation activities is rather stable over the sample period. About 21% of firms in the sample report product innovation activities, ranging from 20% in 1990 to 21% in 1998. About 27% of firms have engaged in process innovation, with the proportion ranging from 27% in 1990 to 30% in 1998. The steady growth in the percentage of exporting firms in our sample corresponds to the change towards openness of the Spanish economy at the beginning of the 1990s. Moreover, exporting activity is characterized by high persistence, owing to the sunk-cost nature of investments in establishing an export position (e.g., Campa, 2004; Roberts & Tybout, 1997). The persistence in exports is also reflected in the transition probabilities shown in Table 2, which reports the average probabilities of keeping/changing the export (or innovation) status in two subsequent years across the sample period. Once a firm has changed its export status the probability of staying in it is about 91%.

The nearly stable percentage of innovating firms in relation to the increasing number of exporters can be explained by the fact that innovation may be needed to *start* export activity, but is not necessarily needed to continue exporting. We then expect the firms to innovate *before* they enter the export market, but not necessarily continue once the entry into exports has occurred. In this case, the percentage of innovating firms stays the same, but these are not the same firms, as they enter and exit innovation activity. Consistent with this, we also find innovation activity being less persistent than exports. The transition probabilities (Table 2) indicate that the probability of dropping innovation activity is about three times higher than the probability of becoming an innovator, given that the firm did not innovate in the previous year. However, the probability of keeping on innovating, given that the firm innovated in the previous year, is about 1.5 times higher than the probability of dropping innovation activity. About 60% of the firms that innovated in the previous year continued innovating in the current year. Thus the transition probabilities suggest that there is a substantial movement in and out of innovation activity.[6]

Empirical Strategy

In this study we are interested in testing the link from product innovation to productivity increase, and ultimately to the decision of a firm to enter the export market. Product innovation may have two effects on the firm's decision to start exporting. On the one hand, product innovation activity may enhance a firm's productivity and as a consequence increase the probability of a firm's starting exporting, owing to the increased chances of more productive firms entering the export market. On the other hand, product innovation may have a direct effect on the decision of a firm to start export operations. Firms may, for example, start export operations in search for greater demand for their newly developed products. We are interested in decomposing and then quantifying the effects of innovation and productivity on exports into a direct component – the direct effects of productivity and product innovation – and an indirect component – the indirect effect of product innovation running through the increased productivity.

Empirically, our argument suggests that product innovation might have a "moderating" effect on the positive association between productivity and export status. Innovating firms show higher productivity levels and grow faster than non-innovators. At the same time, these firms are more likely than

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non-innovating firms to become exporters. In the empirical setting, omitting an innovation variable from the analysis may lead to overestimation of the productivity – exports association. We thus expect the link between productivity and exports to diminish once we account for previous innovation activity (and product innovation in particular). For non-innovators, however, we expect to find a stronger positive association between exporting and productivity.

Key variables

The mechanism that we aim to test in this study implies a particular direction of causality, running from product innovation to productivity and finally to the decision to start exporting. Ultimately, we are interested in the effect of product innovation and productivity on the decision of a firm to export.

Thus our dependent variable is the export status of a firm, and our main independent variables are product innovation status and productivity.[7]

The ESEE survey contains detailed information on exporting activities for every year. We define exporters as firms exporting in year t, that is, with positive sales outside Spain, and non-exporters as firms that did not report any exporting activity in year t. So the dependent variable is a dichotomous variable that equals 1 if a firms reports positive sales outside Spain in year t, and 0 otherwise. The data also include yearly information on the innovation activities of a firm. We define firms as product innovators if they report a product innovation in a particular year. The process innovators are defined in a similar way.[8] We use a dummy variable *Product innovation* measured in year $t-1$ as our main independent variable.[9]

To measure firm productivity we use total factor productivity (TFP). The ESEE data provide information on the output and input variables needed to measure TFP at the firm level. We construct an index of TFP using a multilateral index developed by Caves, Christensen, and Diewert (1982) and extended by Good, Nadiri, and Sickles (1997). Our method of computing the TFP index is similar to that used in recent papers by Aw et al. (2001), Aw, Chung, and Roberts (2000), Aw, Roberts, and Winston (2007) and Delgado et al. (2002). The main advantage of this productivity measure is that the parameters of the production function need not be estimated to compute productivity (Delgado et al., 2002); thus, this type of productivity measure offers the robustness of being a nonparametric first-order approximation to a general production function (Syverson, 2010).[10] The index is computed as each firm's output and input deviations from a reference firm. The reference firm is a hypothetical average firm that varies across industries, that is, each firm's output, inputs and productivity for each year are measured relative to this hypothetical firm in the same industry. Firms with positive residuals are relatively more productive, and firms with negative residuals relatively less productive. We calculate the index as the logarithm of the firm's output minus a cost-weighted sum of the logarithms of the three inputs in the production process (labor, capital and raw materials). The TFP index thus reflects all those factors (other than labor, capital and material inputs) not explicitly taken into account but which contribute to the firm's output and can lead to profit differences across firms. The computation procedure is described in detail in the Appendix. We measure our productivity variable at time t.

Statistical approach

We start by identifying the moderating effect of product innovation on the productivity – export association. To do so, we first test the link between exports, product innovation and TFP for the whole sample of firms. Then we divide the sample into two subsamples based on the firm's previous innovation status: the firms with no product innovation and the firms with product-innovation activity. We proceed by decomposing the effect of product innovation into two channels – the direct effect of innovation on the decision to export, and an indirect effect of product innovation running through the productivity channel.

We estimate a set of three models. Model 1 links the firm's productivity and its determinants. Following our hypothesis, we include the product innovation decision as a main independent variable. Based on prior studies, we include process innovation (e.g., Huergo & Jaumandreu, 2004), import status (e.g., Amiti & Konings, 2007), size, foreign capital and macroeconomic effects approximated by the variation in the gross domestic product (GDP) of Spain as other possible productivity determinants.

$$\mathrm{TFP}_{it} = f(\text{Product innovation}_{it-1}, Z_{it-1}; \gamma^{\mathrm{TFP}}, \beta^{\mathrm{TFP}}) \quad (1)$$

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Models 2 and 3 link our dependent variable, the export dummy, with the product innovation and the TFP variables:

$$\text{Export}_{it} = f(\text{Product innovation}_{it-1}, \text{TFP}_{it}, X_{it}; \gamma, \beta) \quad (2)$$

$$\text{Export}_{it} = f(\text{Product innovation}_{it-1}, Z_{it-1}; \gamma, \beta) \quad (3)$$

where Z_{it-1} are the determinants of the TFP level from Eq. (1).

Following prior research, we include in models 2 and 3 a number of relevant variables that may have an effect on the decision to adopt export activities. Firms' size (*Size*), measured as the logarithm of the number of employees, stands as an important control variable that may affect the export decision. Prior literature posits a nonlinear relationship between size and exporting, showing the positive effect of size up to a certain threshold (Bernard & Jensen, 1999; Sterlacchini, 1999). As our focus is on SMEs, we expect a positive effect of size on the decision to export. Being part of a foreign company might facilitate the process of becoming an exporter (Basile, 2001); thus we include a dummy variable that controls for foreign ownership of the firm (*Foreign capital*). The expansive or recessive states of the market and economy may also exert a positive/negative effect on firms' decision to start operating abroad. Two dummy variables, *Expansion* and *Recession*, control for that eventuality. Finally, we include the vectors of industry and year dummies to control for the industry heterogeneity and macroeconomic conditions common to all firms.

By combining the coefficients of TFP and product innovation in models 1–3 we can decompose the total effect of product innovation into direct and indirect effects running through productivity increases. The *direct* effect of product innovation on the decision to export can be obtained from the coefficient of *Product innovation* in Eq. (2), where both *Product innovation* and *TFP* are included as independent variables. The *total* effect of product innovation can be obtained from the estimate of *Product innovation* in model 3. An indirect effect of product innovation arises when firms are more likely to start exporting owing to higher productivity (as follows from Eq. (2)), *and* product innovation affects the TFP level of firms (as follows from Eq. (1)). Therefore, we obtain the *indirect effect through productivity* by combining the coefficient of *Product innovation* in model 1 and the coefficient of *TFP* in model 2.

To test for the moderating effect of product innovation on the productivity – exports association further, we perform a split-sample regression by distinguishing product innovators from firms with no product innovation. We define the latter as those firms that did not have product innovation in year $t-1$. We define the former as firms that obtained product innovation in year $t-1$. We estimate Eq. (2) across these subsamples of firms. If product innovation affects positively both the productivity level of a firm and the decision to export, we expect to find more productive and export-oriented firms in the group of product innovators, and therefore a diminished link between TFP and exports.

Endogeneity concerns

An issue of endogeneity in the relation between product innovation, productivity and exports deserves particular attention. Biases might result from omitted variables that affect both export and innovation choices and productivity levels: thus, estimating unbiased coefficients requires econometric methods that account for omitted variables (Angrist, 2001; Hamilton & Nickerson, 2003). There might also exist a reverse causal effect from exports to productivity and innovation output – the so-called learning-by-exporting effect documented by empirical studies (e.g., Golovko & Valentini, 2009; Salomon & Jin, 2008; Salomon & Shaver, 2005).

We attempt to control for possible estimation biases in the export decision regression in several ways. Our first step is to take advantage of the yearly panel structure of our data and control for time-invariant unobserved firm heterogeneity. We choose the random effects specification to handle the unobserved factors, owing to the dichotomous nature of the dependent variable. More specifically, we use a dynamic random effects regression to model the decision to export. Prior research has shown that export activity is characterized by high persistency, owing to the sunk costs that it entails (e.g., Campa, 2004; Roberts & Tybout, 1997). To control for possible persistency in the exporting activity, we include the 1-year lagged value of the dependent variable, which makes our model dynamic. To account for the dynamic panel structure we apply the procedure described in Wooldridge (2005). Wooldridge (2005) suggests including the initial

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dependent variable values together with all lags of strictly exogenous covariates.

To address the reverse causality issue, and in support of the findings of the regression analysis, we conduct an additional test with a matching estimator. To sort out the direction of causality between product innovation, productivity and exports, the ideal experiment would be to compare the innovation and productivity of a firm after it started exporting with the innovation and productivity of the same firm had it never exported, provided that a firm engaged in exporting. However, we cannot observe the counterfactual outcome – what would have happened if a firm had not exported provided that a firm engaged in exporting. *Matching estimators* may provide an interesting solution to this problem.[11] In short, matching estimators impute the missing potential outcomes of treated individuals (i.e., what would have happened had the treatment not occurred) by using the outcomes of individuals with *similar* values of "relevant" pre-treatment variables or covariates, but which were not exposed to treatment (i.e., using the outcome of untreated individuals that were similar to treated individuals before the treatment along a number of appropriate dimensions).[12] Several matching estimators exist, see Imbens (2004) for a review. In this study, we use the matching estimator developed by Abadie and Imbens (e.g., Abadie, Drukker, Herr, & Imbens, 2001; Abadie & Imbens, 2002).[13] We include as relevant matching covariates (i.e., as variables relevant to explain the decision to enter into foreign markets) the control variables from the regression analysis: firm size measured as the logarithm of the number of employees; a foreign capital dummy; and year dummy. We also add additional matching variables that may affect (even indirectly) the decision of a firm to export, in order to increase the quality of the "match" between non-exporting firms and firms entering into exports. These are a dummy variable that indicates firm's import status, and a dummy variable that indicates whether the firm carries out R&D.

Using a matching estimator will allow us to trace the changes in product innovation and productivity before and after entry into the export market. We examine the relationship between productivity and exports and product innovation and exports for firms *entering the export market*, compared with matched firms not entering the export market. As a result, we avoid mixing the effects of always exporting firms (established exporters) with the entrants into exports. An increase in productivity and innovation before the entry into exports would be in line with our hypothesized mechanism.

RESULTS

Table 3 presents summary statistics and product moment correlations for our sample.

Generally, the descriptive statistics and correlations are as expected. Export and product innovation indicators show positive and significant correlation with the productivity measure (r=0.12, $p<0.001$ and r=0.07, $p<0.001$, respectively) in line with prior research. Additionally, export and product innovation decisions are positively correlated (r=0.20, $p<0.001$). The decision to innovate in process is also positively and significantly correlated with export and productivity (r=0.13, $p<0.001$ and r=0.01, $p<0.10$, respectively), with the correlation coefficient being substantially lower than in the case of product innovation. This is in line with our hypothesis on the importance of product innovation in explaining both productivity improvements and the decision to start exporting.

Table 3 Descriptive statistics and product moment correlations

	1	*2*	*3*	*4*	*5*	*6*	*7*
1. Export	1						
2. Total factor productivity level	0.12	1					
3. Product innovation	0.21	0.07	1				
4. Process innovation	0.12	0.02	0.27	1			
5. Import	0.49	0.13	0.18	0.12	1		
6. Size (number of employees)	0.39	0.04	0.13	0.16	0.41	1	
7. Foreign capital	0.27	0.08	0.05	0.06	0.29	0.38	1
Mean	0.43	−0.08	0.21	0.27	0.43	45	0.10
Standard deviation	0.49	0.23	0.40	0.44	0.49	48.9	0.30
Min	0	−3.33	0	0	0	1	0
Max	1	1.33	1	1	1	200	1

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Regression Results

We then proceed by estimating models 1–3, which decompose the effect of product innovation into the direct effect on the decision to export, and an indirect effect going through productivity, and test for the moderating effect of product innovation on the exports – productivity link. Table 4 presents the results of the estimation of the productivity regression, and Table 5 reports the results of the export decision regression.

The results of the random effects panel model in Table 4 show that, consistent with the proposed mechanism, product innovation has a positive effect on the productivity level of a firm, significant at the 10% significance level. With respect to the control variables, the decision to innovate in process does not affect the future TFP level.[14] Previous import status of a firm, consistent with our expectations, has a positive and significant effect on the future productivity. Firm size has a negative and significant effect on the productivity level, which may point to the existence of a curvilinear relationship between productivity and size. Foreign capital in the ownership structure of the firm affects the productivity positively and significantly. Finally, the positive change in GDP of Spain is also associated with an increase in the productivity level.

Table 4 The determinants of the TFP level: random effects model

	Model 1
Product innovation ($t-1$)	0.01* (0.005)
	[0.01]
Import ($t-1$)	0.03*** (0.006)
Size	−0.02*** (0.004)
Foreign capital	0.04*** (0.01)
Process innovation ($t-1$)	−0.001 (0.004)
Change in GDP (dGDP)	0.003*** (0.001)
	7809
	Wald $\chi^2(6)=86.58$
	Prob$>\chi^2=0.000$

Standard errors are given in ( ). Marginal effects are given in [ ].
*, ** and *** are significantly different from zero at the 10%, 5% or 1% level, respectively.

Table 5 The export decision regression: dynamic random effects probit model

	Direct effect of product innovation (model 2)	*Total effect of product innovation (model 3)*	*Moderating effect of product innovation*	
			Subsample of product innovators	*Subsample of non-innovators*
Export ($t-1$)	1.77*** (0.06)	1.66*** (0.06)	2.25*** (0.14)	1.89*** (0.07)
Export (0)	1.34*** (0.09)	1.29*** (0.09)	0.91*** (0.18)	1.17*** (0.11)
TFP (t)	0.32*** (0.12)		0.09 (0.23)	0.35*** (0.13)
	[0.12]			
Product innovation ($t-1$)	0.18*** (0.06)	0.20*** (0.06)		
	[0.07]	[0.08]		
Import ($t-1$)		0.46*** (0.05)		
Process innovation ($t-1$)		0.02 (0.05)		
dGDP		0.05*** (0.01)		
Size	0.34*** (0.03)	0.29*** (0.03)	0.28*** (0.07)	0.32*** (0.04)
Foreign capital	0.39*** (0.11)	0.28** (0.10)	0.04 (0.21)	0.46*** (0.12)
Expansive market	0.17*** (0.05)	0.17*** (0.06)	0.32** (0.13)	0.12* (0.06)
Recessive market	−0.02 (0.06)	−0.04 (0.06)	−0.29**(0.14)	0.03 (0.07)
Intercept	−3.25*** (0.30)	−2.89*** (0.26)	−1.67*** (0.60)	−2.44*** (0.28)
Industry and time effects	Yes	Yes	Yes	Yes
	7805	8748	1622	6183
	Wald $\chi^2(34)=2548.19$	Wald $\chi^2(35)=2806.92$	Wald $\chi^2(33)=465.28$	Wald $\chi^2(33)=2055.61$
	Log likelihood=−2023.88	Log likelihood=−2241.28	Log likelihood=−403.89	Log likelihood=−1624.15
	Prob$>\chi^2=0.000$	Prob$>\chi^2=0.000$	Prob$>\chi^2=0.000$	Prob$>\chi^2=0.000$

Direct effect of product innovation: Marginal effect of product innovation (model 2): 0.07***.
Total effect of product innovation: Marginal effect of product innovation (model 3): 0.08***.
Indirect effect of product innovation: Marginal effect of TFP (model 2): 0.12***; Marginal effect of product innovation (model 1): 0.01*.
Standard errors are given in ( ). Marginal effects are given in [ ].
*, ** and *** are significantly different from zero at the 10%, 5% or 1% level, respectively.

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Table 5 reports the results of the estimation of models 2–3, in which we decompose the effect of product innovation into the two distinct channels, and test for the moderating effect of product innovation. The first column in Table 5 lists the estimation results for Eq. (2). Both product innovation and productivity are positively associated with the export status of a firm. The coefficients of *TFP* and *Product innovation* are positive and highly significant ($p<0.01$). With respect to the control variables, previous export activity is positive and significant ($p=0.001$) in determining the current export status, confirming the results of the prior literature. Firm *size* has a positive and significant ($p=0.001$) impact on the decision to export. The *Foreign capital* dummy coefficient shows the expected sign (positive) and a high significance level, suggesting, in line with prior research, that foreign-owned firms are more likely to become exporters. The expansion state of the demand is an important factor affecting the decision of a firm to start exports, with the expected positive sign of the coefficient. The estimate of the *Recession* dummy has the expected but statistically insignificant sign (negative).

We now turn to the direct vs indirect effects of product innovation on the export decision. The direct effect is obtained from the *Product innovation* coefficient in column 1 (model 2). The parameter estimate of product innovation is positive and significant at the 1% significance level, suggesting that carrying out product innovation in a previous year directly increases the probability of exporting by approximately 7%. The total effect of product innovation is provided by the *Product innovation* coefficient in column 2 (model 3). As expected, the total effect of product innovation is also positive and significant, showing that the chances of entering the export market increase by 8% if a firm had product innovation in the previous period.

The existence of the indirect effect of product innovation through productivity is supported by the results of models 1 and 2. The coefficient of *TFP* in column 1 (model 2) shows that firms with a higher productivity level are significantly more likely to start exporting. Furthermore, from the productivity regression (model 1, Table 4) we can see that product innovation in the previous period has a positive and significant (at the 10% level) effect on the next period productivity level.

The indirect impact of the product innovation variable can be obtained by considering the total and direct effects of product innovation. Comparison of the magnitudes of the total and direct effects suggests that product innovation indirectly accounts for about a 1% increase in the probability of exporting. The result is also in line with the productivity regression, where the effect of previous product innovation amounts for about a 1% increase in the productivity level. In such a way, we find evidence of the existence of a significant and positive indirect channel running through productivity increase. The findings, however, point to the prevalence of the direct impact of product innovation on the decision to export.

The power of our argument is only really appreciated when we examine the results of the split sample. Columns 3 and 4 of Table 5 list the results of the export regression for the two subsamples of firms depending on the previous innovation status of the firm. We find support for the existence of the moderating effect of product innovation on the export – productivity association. The coefficient of *TFP level* is positive and highly significant in the group of firms with no product innovation in the previous year. The results are drastically different for the subsample of product innovators. The coefficient of the *TFP* variable is positive but not statistically significant any more, that is, the effect of TFP on exports has vanished once we control for product innovation.

Matching Results

To control for possible reverse causality in the relationship between exports, innovation and productivity, and in order to find confirmation for the results of the panel regression, we then use a matching procedure to estimate the difference in the likelihood of product innovation and the level of TFP between matched exporting and non-exporting firms around the time of entry into exports.

Table 6 lists the results of the matching procedure.

Overall, the results are consistent with those of the regression analysis. Future exporters show a significant increase both in the average productivity level (row 1) and in product innovation frequency (row 2), compared with similar non-exporters already 1 year before entering the export market. This may indicate that future exporters start innovating *before* exporting in order to enter into the export market with new or modified products. At the same time, these firms also

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Table 6 Matching estimator: differences in product innovation and TFP levels between matched exporting and non-exporting firms by years before/after entry into exports.

	3 years before	*2 years before*	*1 year before*	*Year of entry*	*1 year after*	*2 years after*	*3 years after*
TFP level	0.062***	0.023	0.049***	0.014	0.050***	0.062***	0.026
	(0.019)	(0.020)	(0.015)	(0.015)	(0.015)	(0.017)	(0.024)
	N=3355	*N*=3410	*N*=3502	*N*=3531	*N*=3461	*N*=3399	*N*=3302
Product innovation	0.035	0.028	0.054**	0.083***	0.052*	0.064*	0.010
	(0.030)	(0.026)	(0.024)	(0.023)	(0.027)	(0.035)	(0.039)
	N=3355	*N*=3410	*N*=3502	*N*=3531	*N*=3461	*N*=3399	*N*=3302
TFP level: subsample of firms with product innovation in a previous year	0.081*	0.059	0.056	−0.048	0.063	0.011	0.086*
	(0.048)	(0.036)	(0.036)	(0.034)	(0.039)	(0.035)	(0.049)
	N=360	*N*=376	*N*=387	*N*=402	*N*=401	*N*=390	*N*=364
TFP level: subsample of firms with no product innovation in a previous year	0.063***	0.017	0.058***	0.033**	0.057***	0.080***	0.025
	(0.021)	(0.021)	(0.018)	(0.016)	(0.017)	(0.020)	(0.028)
	N=2732	*N*=2762	*N*=2824	*N*=2876	*N*=2807	*N*=2756	*N*=2685

Standard errors are given in parentheses.
*, ** and *** are significantly different from zero at the 10%, 5% or 1% level, respectively.

increase their productivity levels. The observed pattern is in line with the hypothesized effect of product innovation on productivity and the decision to export.

Next, we divide the firms into two subsamples based on previous product innovation activity. The results show no significant difference in productivity levels between entrant firms and non-exporting firms in the group of firms with product innovation, which confirms the findings of the regression analysis. On the other hand, we observe a significant difference in productivity between future exporters and non-exporters for the non-innovating firms, again in line with the regression results.

Overall, our findings show that, once we account for the product innovation carried out in the previous year, the positive association between productivity and export status of a firm vanishes. However, the positive effect of productivity on exports persists for the group of non-product-innovating product firms. Taken together, these findings suggest that previous product innovation activity is partly responsible for the productivity improvements, and consequently the decision to enter into exports.

Robustness and Sensitivity Analysis

To assess the robustness and sensitivity of our results, we tested several other specifications of our main model. To start with, we ran the analysis for different lag structures of the product innovation variable. We report the results of the test for the moderating effect of product innovation in Table 7.[15]

The results confirm our findings. Both product innovation and TFP level have a positive and significant effect on the decision to export, with the coefficients being significant at the 1% significance level. Once we divide the sample into two subsamples, depending on the innovation status of firms, we observe a significant positive association between productivity and exports in the group of firms with no product innovation. In the group of innovators, however, the coefficient of TFP level is positive but not significant, confirming our previous findings.

The next specification uses an alternative measure for product innovation. To eliminate the possible confounding effect of process innovation, we define a firm as a product innovator if it had a product innovation in year $t-1$ exclusively, that is, a firm had a product but not process innovation in year $t-1$. Table 8 summarizes the results of the regression estimation.

Again, the results are in accordance with our main findings. We confirm the moderating effect of product innovation on the export – productivity association (columns 1–2 of Table 8). The positive significant effect of TFP level on the export decision is observed in the subsample of firms with no innovation, but not in the subsample of firms with product innovation defined exclusively.

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Table 7 Dynamic random effects probit model: moderating effect of product innovation. Product innovation: at least once in 2 years

	Whole sample	*Subsample of product innovators*	*Subsample of non-innovators*
Export ($t-1$)	2.04*** (0.08)	2.39*** (0.12)	2.14*** (0.07)
Export (0)	1.04*** (0.12)	0.81*** (0.18)	0.82*** (0.11)
TFP (t)	0.39*** (0.12)	0.19 (0.22)	0.41*** (0.13)
Product innovation ($t-1$) or ($t-2$)	0.23*** (0.06)		
Size	0.28*** (0.04)	0.30*** (0.07)	0.23*** (0.04)
Foreign capital	0.26** (0.11)	0.07 (0.19)	0.28** (0.11)
Expansive market	0.21*** (0.06)	0.38*** (0.12)	0.11 (0.07)
Recessive market	−0.10 (0.06)	−0.25* (0.13)	−0.05 (0.07)
Intercept	−2.31*** (0.29)	−2.57*** (0.64)	−1.96*** (0.26)
Industry and time effects	Yes	Yes	Yes
	6304	1850	4454
	Wald $\chi^2(33)=2330.76$	Wald $\chi^2(32)=557.99$	Wald $\chi^2(32)=1940.95$
	Log likelihood=−1624.07	Log likelihood=−444.32	Log likelihood=−1167.16
	Prob > χ^2=0.000	Prob > χ^2=0.000	Prob > χ^2=0.000

Standard errors are given in parentheses.
*, ** and *** are significantly different from zero at the 10%, 5% or 1% level, respectively.

Table 8 Dynamic random effects probit model: moderating effect of product innovation. Product innovation defined exclusively

	Whole sample	*Subsample of product innovators*	*Subsample of non-innovators*
Export ($t-1$)	1.96*** (0.08)	2.51*** (0.21)	2.01*** (0.08)
Export (0)	1.06*** (0.11)	0.44* (0.23)	0.99*** (0.13)
TFP (t)	0.28*** (0.12)	0.25 (0.30)	0.27** (0.13)
Product innovation ($t-1$) excl.	0.25*** (0.08)		
Size	0.31*** (0.04)	0.39*** (0.10)	0.28*** (0.04)
Foreign capital	0.45*** (0.12)	0.06 (0.31)	0.50*** (0.13)
Expansive market	0.17** (0.06)	0.36** (0.17)	0.13* (0.07)
Intercept	−2.18*** (0.28)	−2.33*** (0.89)	−3.12*** (0.37)
Industry and time effects	Yes	Yes	Yes
	5854	818	5036
	Wald $\chi^2(33)=1985.71$	Wald $\chi^2(32)=189.90$	Wald $\chi^2(32)=1699.79$
	Log likelihood=−1533.45	Log likelihood=−218.04	Log likelihood=−1305.24
	Prob > χ^2=0.000	Prob > χ^2=0.000	Prob > χ^2=0.000

Standard errors are given in parentheses.
*, ** and *** are significantly different from zero at the 10%, 5% or 1% level, respectively.

DISCUSSION AND CONCLUSION

Extant international trade literature has addressed the positive association between exporting and firm performance, and productivity in particular. Empirical studies mostly agree on the prevalence of the selection mechanism over the effect of learning in explaining this correlation. The literature, however, does not explain which factors lead these more productive firms to gain productivity advantages that allow them to enter the exports market. In this study we address this gap, and examine the drivers of productivity heterogeneity that can explain the well-documented link between productivity and exports. We draw on the industrial dynamics, international trade and international business literatures to argue that a firm's prior decision to innovate in product may partially explain the productivity – exports association.

Our findings emphasize that product innovation has an important moderating effect on the positive association between exports and productivity.

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Once we control for the previous innovation status of a firm, this firm's productivity turns out to be associated less with firm participation in the export market. Moreover, we find empirical evidence for both a direct and an indirect effect that product innovation might have on the decision of a firm to start exporting. Product innovation improves productivity levels, which pushes firms to enter the export markets. Additionally, product innovation directly affects the probability of a firm's starting export operations.

A number of recent empirical studies provide support for the direction of our argument. In a recent paper, Aw et al. (2007) find that, for firms that do not invest in R&D, the productivity of exporters is significantly higher than that of non-exporting firms. Moreover, firms that export and invest in R&D are found to have higher productivity than those that only export. This evidence is used to argue that exports and R&D are complementary for productivity growth, with R&D activities facilitating assimilation of the benefits from export markets. Aw, Roberts, and Xu (2009), using a sample of Taiwanese firms from the electronic products industry, test for the potential links between exports, R&D investment and productivity. They find a positive relationship between investing in R&D and exporting and future productivity distributions, with the effect of R&D being more pronounced. They also observe the self-selection of high-productivity firms into exports and R&D.

Interestingly, the positive link between productivity and exports observed in prior research does continue to exist for firms with no product innovation. This result might be consistent with the existence of other types of productivity-enhancing investments emphasized by recent studies (e.g., Bellone, Guillou, & Nesta, 2009; Lileeva & Trefler, 2010). For instance, Bellone et al. (2009), using a sample of French firms, find that innovation is positively correlated with productivity, but cannot account for all the variance in productivity between exporting and non-exporting firms. Their results show that exporters are more productive than non-exporters even if they do not innovate, arguing that export activity may require the implementation of new competencies. These new competencies, together with organizational changes, may be responsible for future productivity improvements.

The positive association between exports and productivity observed in the group of firms with no product innovation might also be consistent with the existence of learning-by-exporting emphasized by recent studies (e.g., Golovko & Valentini, 2009; Salomon & Jin, 2008), which could be particularly relevant for this group of firms. The observed superior productivity of exporting firms may be related to their ability to get new technological knowledge in the exports markets, and to the tougher competition abroad. The results of the matching estimation provide support for the possible learning-by-exporting effect, as we also observe increased productivity *after* the entry into exports.

Our paper contributes to the literature in several ways. By bringing together and integrating the elements of three different research areas, we highlight the distinct effects that innovation has on the decision to start exporting. The general focus of the international business literature has been mainly on FDI, the motives to do FDI and the optimal mode of entry into new markets at a particular point in time (such as through licensing, greenfield investment, acquisition or joint venturing: e.g., Buckley & Casson, 1976; Dunning, 1980). Exporting, being the most widely used firm strategy for international expansion, has received less attention. This study adds to our still scant understanding of export strategies. In particular, our results shed some new light on a stream of research that explains the antecedents of the decision to export. Prior empirical research in this area has highlighted the importance of product innovation for the decision to export (Cassiman & Martinez-Ros, 2007). We take a different perspective, and provide evidence on the existence of two possible channels for the effect of product innovation on the decision to start exporting.

Moreover, we contribute to research on productivity and firm-level exports by empirically showing the importance of prior firm's investments for explaining the self-selection mechanism of more productive firms into exports. Hence, our findings highlight the importance of firm decisions for their future growth paths, and for explaining performance heterogeneity among firms.

Our results also present some important implications for practice. Our finding on the role of product innovation in driving productivity and exports appears to be relevant from a public policy perspective. Economic policies aimed at export promotions have been widely supported by the argument that exposing firms to international markets through exports leads to increased productivity of exporters. However, empirical evidence

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shows that these efforts may not work as well as thought, if not combined with important innovation efforts by the firms. As a result, and despite large investments made by the governmental agencies in export promotions during the 1990s, the productivity growth of Spain has been one of the lowest among the EU countries. Moreover, even if the policy works in the right direction, it may not necessarily be optimal. Our findings show that innovation activity – and, more specifically, product innovation – may be responsible for both productivity enhancements and entry into exports. Then policies aimed at innovation promotion, and product innovation in particular, might be an alternative to direct export promotions. To make a proper comparison of various policy alternatives, however, one would need to account for many other factors, not least the cost of these policies. Nevertheless, as Syverson (2010) notes, research has typically compared the effects of policy reforms with a null of no reform, but perhaps an equally important comparison is among possible policy alternatives, such as innovation vs export promotion.

There are several other avenues for future research suggested by our findings. The results show that the link between exports and productivity persists for the non-innovating firms. Focusing on the subsample of non-innovators, thus sorting out the confounding effect of innovation may be a "cleaner" setting for a test of selection vs the learning-by-exporting hypothesis in the productivity – exports link. Another interesting extension of our work could examine alternative productivity-enhancing investments and their interplay with investing in innovation. We have focused on innovation as one such productivity-enhancing activity. Recent research has started to look at several other activities, for example, investing in imports (e.g., Lileeva & Trefler, 2010). Product innovation and imports may serve as productivity-enhancing investments, allowing firms to enter the export market afterwards. As firms import higher-quality inputs, new materials and components, they are able to transform them into higher-quality outputs while developing new and better products (e.g., Iacovone & Javorcik, 2008; Kugler & Verhoogen, 2008). We should therefore expect imports, or firm internationalization on the input side, to be highly related to innovation and, as a consequence, to productivity improvements. Moreover, innovation, imports and exports may be complementary activities in their effect on productivity, that is, exporting may actually enhance the returns to such investment decisions.

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NOTES

[1]Recent studies report evidence in favor of learning-by-exporting. De Loecker (2007), for a sample of Slovenian firms, finds that export entrants become more productive once they start exporting. Van Biesebroeck (2005) finds evidence of exports increasing firm productivity for a sample of sub-Saharan African firms. Salomon and Shaver (2005) and Salomon and Jin (2008), for a sample of Spanish firms, find that exporting is associated with the post-entry increase in patent applications.

[2]We consider the self-selection mechanism of more productive firms into exports, confirmed by most of the empirical studies, as the main explanation for the correlation between productivity and exports. However, we do not exclude the possibility of learning-by-exporting. Our assumption is that the effect of learning-by-exporting is less pronounced than that of self-selection, which has also been confirmed by most of the empirical findings. Nevertheless, in the discussion we argue that a focus on non-innovating firms might be more appropriate to uncover evidence from learning-by-exporting.

[3]The proportion of the firms in year t that continue in the survey in year $t+1$ is approximately 90% for the 1990–1998 sample period. Among the firms that exited the sample, approximately 2.2% disappeared and approximately 7.7% stopped collaborating.

[4]A less restrictive definition of foreign-owned companies will also include firms with more than 10% and less than 50% of foreign capital. Only about 1.5% of SMEs in our sample are with foreign ownership between 10% and 49% of foreign capital.

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[5]Pla-Barber and Escriba-Esteve (2006) show the existence of a group of firms that follow a relatively faster internationalization process compared with the traditional sequential internationalization. These firms begin to export at an early stage (1–6 years from founding) and devote substantial efforts to the international arena, thus complying with the definition of born global firms. Therefore their results suggest that firms from a late-investor country do not necessarily need to follow a slow, sequenced pattern to enter the international markets effectively.

[6]In addition, comparison of the variance measure of export and product innovation variables shows that the variation of innovation activity is higher than that of exports.

[7]Table A1 in the Appendix provides definitions of all the variables used in the analysis.

[8]The ESEE survey uses the following definitions of product and process innovation. For product innovation, the firms were asked whether they obtained a product innovation in a given year. Product innovation was defined as new products, or products with new features that were different from those that a firm had produced in previous years. If the answer to this question was "yes", the type of modification was asked: (1) whether the innovation incorporated new materials; (2) whether the innovation incorporated new components or intermediate products; (3) whether the innovation incorporated new design or presentation; and (4) whether the new product performed new functions. For process innovation, the question was whether a firm introduced an important modification in the current production process. If the answer was "yes", the type of product innovation was asked: (1) whether it was related to the introduction of the new machinery; (2) whether it was related to the introduction of new methods of production organization; and (3) whether it was both types of process innovation.

[9]As innovation does show some persistence, we experimented with various lag specifications. Our basic result is robust to these alternative specifications. See the "Robustness and sensitivity analysis" section.

[10]Syverson (2010) indicates that the empirical facts on productivity documented in the literature show little sensitivity to the actual choice of productivity measures. Van Biesebroeck (2007) argues that the index measures of productivity are well suited for the type of analysis we perform.

[11]Typically popular in the labor economics literature, this non-parametric technique has been also used by MacGarvie (2006) to test the direction of causality between exporting, importing and innovation measured by patent citations, and by De Loecker (2007) to examine the causal link between exports and productivity.

[12]What makes a variable "relevant" and appropriate is the extent to which the variable affects the probability of being subject to treatment.

[13]We estimate a matching estimator using a STATA routine developed by Abadie et al., 2001.

[14]This result needs careful interpretation. Process innovation usually involves changes in the production process aimed at improving production efficiency, thus presumably having a direct effect on firm's productivity. We do not observe such an effect in our data. The possible explanation could be that, very often, process innovation is incorporated in the capital investment, as suggested by Crespi, Criscuolo, and Haskel (2007): process innovation "is a combination of advanced capital investment and process organizational change". As we already control for capital investment in our productivity measure (TFP), we should not expect a significant effect of process innovation on productivity measured as TFP.

[15]We have tested the model for the different lag structures of product innovation: product innovation at least once in two previous years (Table 8); product innovation in both year $t-1$ and $t-2$; product innovation in year $t-2$; and product innovation once in 3 years. Our finding on the moderating effect of product innovation on the export – productivity association is robust to these alternative specifications. We report only the results for the product innovation measured as innovating at least once in the two previous years.

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APPENDIX

See Table A1.

Table A1 Variable definitions

Variable	*Description*
Export	Export status dummy, equal to 1 if firm exports at time *t*, and 0 if it performs no exporting activities at time *t*
TFP level	Firm-specific index of TFP, constructed using a multilateral index (see below for a detailed explanation of its calculation)
Product innovation	Dummy variable, equal to 1 if firm carried out product innovation at time $t-1$, and 0 if firm performed no product innovation at time $t-1$
Size	Firm size, measured as the logarithm of the number of employees
Foreign capital	Foreign capital dummy, equal to 1 if firm has more than 50% of foreign capital at time *t*
Import	Dummy variable, equal to 1 if firm had imports at time $t-1$, and 0 if firm had no imports at time $t-1$
Process innovation	Dummy variable, equal to 1 if firm carried out process innovation at time $t-1$, and 0 if firm performed no process innovation at time $t-1$
R&D investments	Dummy variable, equal to 1 if firm invested in R&D at time $t-1$
Expansion	Dummy variable, equal to 1 if market evolution with respect to the previous year was perceived by a firm as expansive
Recession	Dummy variable, equal to 1 if market evolution with respect to the previous year was perceived by a firm as recessive
dGDP	Change in the GDP of Spain at time t compared with time $t-1$
Definitions of product and process innovation in the ESEE survey:	
Product innovation	Whether a firm obtained product innovation in a given year – new products, or products with new features that are different from those that a firm produced in the previous years. If the answer is yes, the type of modification is asked: • incorporates new materials • incorporates new components or intermediate products

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Table A1 *Continued*

	• incorporates new design or presentation • the product performs new functions
Process innovation	Whether a firm introduced an important modification in the production process. If the answer is yes, the type of modification is asked: • introduction of new machinery • introduction of new methods of production organization • both

Calculation of the Total Factor Productivity (TFP) Index

The productivity index is computed as a multilateral index, developed in Caves et al. (1982) and extended in Good et al. (1997). In calculating the TFP index we follow Aw et al. (2000, 2007) and Delgado et al. (2002). The TFP index measures the proportional difference in *TFP* for firm *i* in year *t* relative to a hypothetical reference firm in the same industry *r*, that is, each firm's output, inputs and productivity for each year are measured relative to this hypothetical firm in the same industry. This reference firm is defined as follows:

(1) The output is equal to the geometric mean of outputs over all observations in industry *r*.
(2) Inputs are equal to the geometric means of inputs over all observations in industry *r*.
(3) Cost shares are the arithmetic means of cost shares over all observations in industry *r*.

Firms with positive residuals are relatively more productive, and firms with negative residuals are relatively less productive We calculate the index as the logarithm of the firm's output minus a cost-weighted sum of the logarithms of the three inputs in the production process (labor, capital and raw materials). More specifically, the TFP index for firm *i* (i=1, ..., N) from industry *r* (r=1, ..., R) in year *t* (t=1, ..., T) is computed using the formula

$$\ln \mathrm{TFP}_{\mathrm{itr}} = \ln Y_{\mathrm{itr}} - \overline{\ln Y_r} - \sum_{j=1}^{J} \frac{1}{2}(S_{\mathrm{itrj}} + \overline{S_{\mathrm{rj}}})(\ln X_{\mathrm{itrj}} - \overline{\ln X_{\mathrm{rj}}})$$

where Y_{itr} is the output of firm *i* in year *t*; X_{itrj} is an input *j* ($j = \overline{1,J}$) of firm *i* in year *t*; S_{itrj} is a cost-based share of input *j* of firm *i* in year *t*; and

$$\overline{\ln Y_r} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} \ln Y_{\mathrm{itr}}$$

$$\overline{\ln X_r} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} \ln X_{\mathrm{itr}}$$

and

$$\overline{S_r} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} S_{\mathrm{itr}}$$

are the same variables for the reference firm.

The ESEE data provide information on the output and input variables needed to measure TFP at the firm level. We model each firm as using three inputs in its production function: labor, capital and material input. The labor input is measured as the number of total effective working hours per year. The measure of capital input is the capital stock, calculated using the formula

$$k_t^* = I_t + k_{t-1}^*(1 - d_t)\frac{P_t}{P_{t-1}}$$

where *I* represents investment in equipment in year *t*, *d* the depreciation rates in year *t*, and *P* the price indexes for equipment in year *t*. The material input includes raw materials, fuel and electricity costs, and other services bought by a firm. The material expenditures are deflated using the firm-specific price indexes for each of the inputs provided in the ESEE survey.

Firm output is defined as total firm sales deflated by a producer price index defined at the two-digit NACE industry level. The information on depreciation rates, price indexes for equipment, producer price index and consumer price index is taken from the Instituto Nacional de Estadística de España (www.ine.es). Cost-based input shares are calculated as the costs of each input in total input costs. The total input cost is the sum of the labor cost, material cost and the cost of capital. Labor costs are measured as total salaries to employees, deflated by the consumer price index. Capital cost is computed using an estimation of the user cost of capital for each firm. User cost of capital is

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calculated as the sum of the cost of long-term debt and depreciation rates less the variation of the price index for capital goods. The cost share of materials is calculated as the residual after subtracting the expenditures on labor and capital.

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